

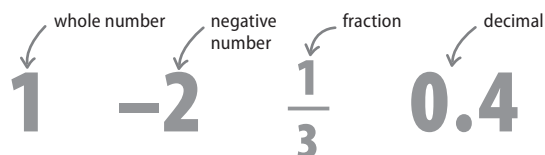
2 Introducing numbers

COUNTING AND NUMBERS FORM THE FOUNDATION OF MATHEMATICS.

Numbers are symbols that developed as a way to record amounts or quantities, but over centuries mathematicians have discovered ways to use and interpret numbers in order to work out new information.

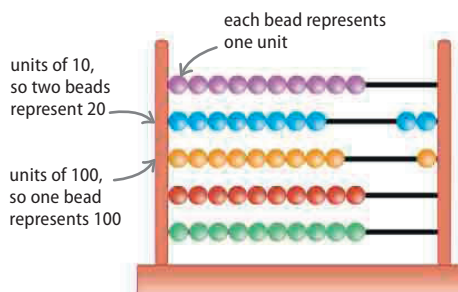
What are numbers?

Numbers are basically a set of standard symbols that represent quantities—the familiar 0 to 9. In addition to these whole numbers (also called integers) there are also fractions (see pp.48–55) and decimals (see pp.44–45). Numbers can also be negative, or less than zero (see pp.34–35).



△ Types of numbers

Here 1 is a positive whole number and -2 is a negative number. The symbol $\frac{1}{3}$ represents a fraction, which is one part of a whole that has been divided into three parts. A decimal is another way to express a fraction.



◁ Abacus

The abacus is a traditional calculating and counting device with beads that represent numbers. The number shown here is 120.

▽ First number

One is not a prime number. It is called the “multiplicative identity,” because any number multiplied by 1 gives that number as the answer.

▽ Even prime number

The number 2 is the only even-numbered prime number—a number that is only divisible by itself and 1 (see pp.26–27).

LOOKING CLOSER

Zero

The use of the symbol for zero is considered an important advance in the way numbers are written. Before the symbol for zero was adopted, a blank space was used in calculations. This could lead to ambiguity and made numbers easier to confuse. For example, it was difficult to distinguish between 400, 40, and 4, since they were all represented by only the number 4. The symbol zero developed from a dot first used by Indian mathematicians to act a placeholder.



zero is important for 24-hour timekeeping

◁ Easy to read

The zero acts as a placeholder for the “tens,” which makes it easy to distinguish the single minutes.



△ Perfect number

This is the smallest perfect number, which is a number that is the sum of its positive divisors (except for itself). So, $1 + 2 + 3 = 6$.

△ Not the sum of squares

The number 7 is the lowest number that cannot be represented as the sum of the squares of three whole numbers (integers).